

PERSONAL INFORMATION

Pais, Matteo

📍 INAF – Astronomical Observatory of Padova, Vicolo dell'Osservatorio 5, 35122

Padova (PD), Italy

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Nationality Italian

WORK EXPERIENCE

- 2024 – present **Postdoctoral researcher**
Istituto Nazionale di Astrofisica (INAF), Astronomical Observatory of Padova, Padova (PD), Italy
Project funded by PRIN **EMERGE** — PI: Dr. Riccardo Ciolfi
- 2020 – 2024 **Postdoctoral researcher**
Hebrew University of Jerusalem (HUJI), Jerusalem, Israel
Projects funded by ERC Advanced Grants **Multijets** and **TReX** — PI: Prof. Tsvi Piran

AFFILIATIONS

- 2024 – present **Postdoctoral research associate**
Istituto Nazionale di Fisica Nucleare (INFN), Padova Division, Padova (PD), Italy
Group IV (Theory) coordinator: Dr. Davide Cassani

EDUCATION AND TRAINING

- 2016 – 2020 **PhD in Physics (Dr. rer. nat.)**
Heidelberg University (MPIA – HITS), Leibniz Institute for Astrophysics Potsdam (AIP), Germany
– Thesis title: *Magneto-hydrodynamic simulation with cosmic rays acceleration at supernova remnants*
– Supervisor: Prof. Christoph Pfrommer
– Co-supervisor: Prof. Ralf Klessen
– Second co-supervisor / thesis committee member: Prof. Volker Springel
– Full text: <https://archiv.ub.uni-heidelberg.de/volltextserver/29011/>
- 2013 – 2016 **MSc in Physics**
University of Cagliari, Italy
– Thesis title: *Inflation as De Sitter Instability*
– Supervisor: Prof. Mariano Cadoni
– Final mark: 110/110 *cum laude*
- 2010 – 2013 **BSc in Physics**
University of Cagliari, Italy
– Thesis title: *Kerr Black Holes*
– Supervisor: Prof. Mariano Cadoni
– Final mark: 110/110 *cum laude*

DOCTORAL AND SUMMER SCHOOLS

- 2018 DIAS Summer School in High Energy Astrophysics, Dublin, Ireland

2017 NBIA School on Plasma Physics, Niels Bohr Institute, Copenhagen, Denmark

AWARDS AND FELLOWSHIPS

- 14/02/2025 **Seal of Excellence** — Marie Skłodowska-Curie Postdoctoral Fellowship
European Commission, Horizon Europe — call HORIZON-MSCA-2024-PF-01-01, proposal 101209980 “KiloJets”
- Four-year postdoctoral fellowship** — Yukawa Institute for Theoretical Physics (YITP), Kyoto, Japan (*offered, declined*)
- 2020 – 2024 **Four-year postdoctoral fellowship** — The Hebrew University of Jerusalem (HUJI), Jerusalem, Israel
- 2016 – 2020 **PhD fellowship** — International Max Planck Research School Heidelberg (IMPRS-HD), Germany

PROFESSIONAL AND ACADEMIC ELIGIBILITY

- 08/2026 Eligibility as INAF Researcher, Level III — “Grandi Progetti” call, Position-06
- 10/2025 Eligibility as INAF Researcher, Level III, fixed-term contract

ALLOCATED COMPUTING RESOURCES

- Research activities in which I act as PI or Co-PI have been awarded a total of **3,700,000 CPU hours** and **64,000 GPU hours**.
- 09/2026 **PI** — ISCRA-C call, project **MAGNUJET**: 64,000 GPU hours
CINECA, ISCRA — Italian SuperComputing Resource Allocation
- 09/2025 **PI** — ISCRA-C call, project **HiNuJetS**: 100,000 CPU hours
CINECA, ISCRA — Italian SuperComputing Resource Allocation
- 11/2025 **Co-PI** — ISCRA-B call, project **MerJet**: 3,600,000 CPU hours (PI: Riccardo Ciolfi)
CINECA, ISCRA — Italian SuperComputing Resource Allocation
- Projects in which I act as collaborator have been awarded over **24,000,000 CPU hours** and **4,000,000 GPU hours**:
- INFN Agreement 2026: 14,000,000 CPU hours — TEONGRAV project (PI: Luca Del Zanna), CINECA — LEONARDO Supercomputer
 - INFN Agreement 2026: 4,000,000 GPU hours — TEONGRAV project (PI: Luca Del Zanna), CINECA — LEONARDO Supercomputer
 - INFN Agreement 2025: 10,000,000 CPU hours — TEONGRAV project (PI: Bruno Giacomazzo), CINECA — LEONARDO Supercomputer

TEACHING ACTIVITY

- 2017 **Teaching assistant** — Ruprecht-Karls-Universität Heidelberg, Germany
MSc course: *General Relativity*
- 2014 **Teaching assistant** — University of Cagliari, Italy
BSc course: *Physics Laboratory*

INSTITUTIONAL DUTIES

- 04/2026 – present **Organiser and co-coordinator** of the weekly astrophysics seminars at INAF – Astronomical Observatory of Padova
- 01/2022 – 07/2023 **Organiser, chair and co-coordinator** of the weekly astrophysics seminars at the Racah Institute of Physics, The Hebrew University of Jerusalem

PEER-REVIEW ACTIVITY

- 2023 – present **Referee** for *The Astrophysical Journal* (AAS) and for MDPI journals

RESEARCH INTERESTS

High-energy astrophysics and plasma physics: cosmic-ray acceleration at shocks, relativistic jets in binary neutron star mergers and in collapsars, multi-messenger transients, high-energy neutrino emission.

Collaborators

- Istituto Nazionale di Astrofisica (INAF), Italy (*Ciolfi, Pavan, Peretti, Amato, Morlino*)
- Sapienza University of Rome, Italy (*Celli*)
- Ruhr University Bochum, Germany (*Zegarelli*)
- Hebrew University, Jerusalem, Israel (*Piran*)
- Ben Gurion University, Beer Sheva, Israel (*Lyubarsky*)
- Tel Aviv University, Israel (*Nakar*)
- Max Planck Institute for Gravitational Physics, Potsdam, Germany — Yukawa Center, Kyoto University, Japan (*Kiuchi, Shibata*)
- Tokyo University, Japan (*Hotokezaka, Irwin*)
- Leibniz Institute for Astrophysics, Potsdam, Germany (*Pfrommer*, during the PhD)

SCIENTIFIC COLLABORATIONS

- 07/2026 – present **KM3NeT** — international collaboration, Naples Unit (Theory)
- 12/2025 – present **MezzoCielo Collaboration** — science team
- 2025 – present **TEONGRAV** — PI: Bruno Giacomazzo, Luca Del Zanna
- 2025 – present **GRAWITA** — Working Group 6, PI: Maria Grazia Bernardini
- 2023 – 2026 **SCEECS** — Simons Foundation, NS Group 2, PI: Roger D. Blandford
- 2022 – 2023 **TCAN** — external collaborator with Piran, PI: Manuela Campanelli

LANGUAGE SKILLS

Mother tongue Italian

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	C1	C1	C1	C1
German	B1	B1	B1	B1	B1
French	A2	B1	A2	A2	A2
Spanish	A2	A2	A1	A1	A1
Hebrew	A1	A1	A1	A1	A1
Japanese	A1	A1	A1	A1	A1

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
[Common European Framework of Reference for Languages](#)

DIGITAL SKILLS

Advanced scientific programming in **Python** and **C/C++**, used daily for the development of numerical codes and for the analysis and post-processing of data and simulations. Advanced command of **LaTeX** for scientific publications and technical documentation. Experience with **HPC** infrastructures, code parallelisation and optimisation for CPU and GPU architectures. Codes used and developed: AREPO, PLUTO, GPLUTO.

ORAL CONTRIBUTIONS TO CONFERENCES AND WORKSHOPS

- 10/2026 Co-author of an oral contribution — *INAF-KM3NeT Workshop*, Caserta, Italy

- 09/2026 Oral contribution — *RICAP 2026*, Rome, Italy
- 04/2026 Oral contribution — *KM3NeT – INAF Initiative*, Rome, Italy
- 03/2026 Oral contribution — *Rencontres de Moriond 2026, VHEPU Conference*, La Thuile, Italy
- 07/2025 Oral contribution — *39th International Cosmic Ray Conference (ICRC 2025)*, Geneva, Switzerland
- 04/2025 Oral contribution — *An Extraordinary Journey into the Transient Sky*, Padova, Italy
- 09/2024 **Invited plenary contribution** — *CNOC XIII, Italian National Congress on Compact Objects*, Alghero, Italy
- 07/2024 **Invited oral contribution** — *17th Marcel Grossmann Meeting*, Pescara, Italy
- 07/2024 Oral contribution — *17th Marcel Grossmann Meeting*, Pescara, Italy
- 07/2024 Oral contribution — *EAS 2024 Annual Meeting*, Padova, Italy
- 12/2023 Oral contribution — *32nd Texas Symposium on Relativistic Astrophysics*, Shanghai, China
- 09/2023 Oral contribution — *TeV Particle Astrophysics (TeVPA 2023)*, Naples, Italy
- 03/2018 Oral contribution — *Workshop on Astrophysical Shocks*, Potsdam, Germany

INVITED SEMINARS

- 05/2026 INAF — Arcetri Astrophysical Observatory, Florence, Italy
- 05/2024 INAF — Astronomical Observatory of Trieste, Trieste, Italy
- 12/2023 Yukawa Institute for Theoretical Physics (YITP), Kyoto, Japan
- 11/2023 INAF — Astronomical Observatory of Padova, Padova, Italy
- 06/2022 Kavli IPMU, Tokyo, Japan
- 10/2021 Astrophysical Seminar, The Hebrew University of Jerusalem, Israel

POSTER CONTRIBUTIONS

- 05/2025 *XV Einstein Telescope Symposium*, Bologna, Italy
- 12/2023 *32nd Texas Symposium on Relativistic Astrophysics*, Shanghai, China
- 02/2023 *Black Holes, from Theory to Observations*, Tel Aviv, Israel
- 12/2022 *Unsolved Problems in Astrophysics*, Jerusalem, Israel
- 06/2019 *Supernova Remnants II*, Chania, Greece
- 09/2018 *Thinkshop on Galaxy Formation*, Potsdam, Germany

PUBLICATIONS

Summary

- **8** peer-reviewed journal articles: **7** as first author, **0** as second author, **1** as third author or later
- **3** contributions in conference proceedings
- **156** citations as of 21/09/2026 — H-index: **7**

Sources: NASA/ADS–SciXplorer and Google Scholar. ORCID: 0000-0002-4243-3889

Peer-reviewed journal articles

1. **M. Pais**, R. Ciolfi, A. Pavan, *Modelling the delayed shock-breakout emission following jet-launching binary neutron star mergers via relativistic magnetohydrodynamic simulations*, *Astronomy & Astrophysics*, 2026, 20 pp. DOI: 10.1051/0004-6361/202661692 — *first author*
2. **M. Pais**, T. Piran, K. Kiuchi, M. Shibata, *Simulating short gamma-ray burst jets in realistic late binary neutron star merger environments*, *The Astrophysical Journal*, 976, 1, id. 35, 2024, 20 pp. DOI: 10.3847/1538-4357/ad7d04 — *first author*
3. **M. Pais**, T. Piran, Y. Lyubarsky, K. Kiuchi, M. Shibata, *The collimation of relativistic jets in post-neutron star binary merger simulations*, *The Astrophysical Journal Letters*, 946, 1, id. L9, 2023, 7 pp. DOI: 10.3847/2041-8213/acc2c5 — *first author*
4. **M. Pais**, T. Piran, E. Nakar, *The velocity distribution of outflows driven by choked jets in stellar envelopes*, *Monthly Notices of the Royal Astronomical Society*, 519, 2, 2023, pp. 1941–1954. DOI: 10.1093/mnras/stac3640 — *first author*
5. G. Winner, C. Pfrommer, P. Girichidis, M. Werhahn, **M. Pais**, *Evolution and observational signatures of the cosmic ray electron spectrum in SN 1006*, *Monthly Notices of the Royal Astronomical Society*, 499, 2, 2020, pp. 2785–2802. DOI: 10.1093/mnras/staa2989 — *fifth author*
6. **M. Pais**, C. Pfrommer, *Simulating TeV gamma-ray morphologies of shell-type supernova remnants*, *Monthly Notices of the Royal Astronomical Society*, 498, 4, 2020, pp. 5557–5573. DOI: 10.1093/mnras/staa2827 — *first author*
7. **M. Pais**, C. Pfrommer, K. Ehlert, M. Werhahn, G. Winner, *Constraining the coherence scale of the interstellar magnetic field using TeV gamma-ray observations of supernova remnants*, *Monthly Notices of the Royal Astronomical Society*, 496, 2, 2020, pp. 2448–2461. DOI: 10.1093/mnras/staa1678 — *first author*
8. **M. Pais**, C. Pfrommer, K. Ehlert, R. Pakmor, *The effect of cosmic-ray acceleration on supernova blast wave dynamics*, *Monthly Notices of the Royal Astronomical Society*, 478, 4, 2018, pp. 5278–5295. DOI: 10.1093/mnras/sty1410 — *first author*

Conference proceedings

1. N. Elias-Rosa, M. Della Valle, M. T. Botticella, E. Cappellaro, R. Ragazzoni, D. Magrin, C. Arcidiacono, **M. Pais**, MezzoCielo Collaboration, *Mezzocielo Telescope: Catching the Earliest Phases of Multimessenger Transients*, *Expanding Horizons 2026*, ESO Garching, 13–17 July 2026, id. 67. DOI: 10.5281/zenodo.21841550
2. S. Celli, **M. Pais**, A. Zegarelli, E. Peretti, *Multi-messenger emission in choked jets from collapsars*, *High Energy Phenomena in Relativistic Outflows IX*, Rio de Janeiro, 4–8 August 2025, id. 11. DOI: 10.22323/1.510.0011 — *second author*
3. **M. Pais**, S. Celli, A. Zegarelli, E. Peretti, *Choked jets in BSG/RSG as possible sources of high-energy neutrinos*, 39th International Cosmic Ray Conference (ICRC2025), Geneva, 15–24 July 2025, PoS(ICRC2025)1138. DOI: 10.22323/1.501.1138 — *first author*